

ALISA OMELCHENKO

347-570-5722 | alisa.a.omelchenko@gmail.com | linkedin.com/in/alisa-omelchenko | [GitHub @AlisaOmel](https://github.com/AlisaOmel)
alisaomel.github.io | Pittsburgh, PA, 15207

RESEARCH PROFILE

PhD candidate in the CMU-Pitt joint Computational Biology program, bridging wet-lab and computational research to build tools that keep biological relevance at the center of methods development. My work spans multi-omic network analysis, protein modeling, and agentic AI applications in immunology and disease, including first-author publication analyzing single-cell transcriptomic and Multiome data, as well as designing machine learning methods. I have hands-on experience evaluating LLM agents for biological research tasks and bring a critical eye to identifying where AI-generated analyses go wrong. Beyond the bench and the laptop, I communicate biological concepts to audiences ranging from bench scientists to policymakers, through mentoring, science policy advocacy, and cross-disciplinary collaboration. I'm drawn to fast-paced environments where I can partner directly with research teams to translate open questions into biological discovery with clinical relevance.

EDUCATION

CMU-Pitt Joint Computational Biology: Pittsburgh, Pennsylvania

PhD Candidate in Computational Biology

Sept 2022- Present

Expected Graduation 2026-2027

NYU Tandon School of Engineering: Brooklyn, New York

M.S. Biotechnology

B.S. BioMolecular Sciences (Magna Cum Laude)

Graduated May 2018

GPA: 3.94

GPA: 3.61

SKILLS

Computer Languages: Python, R, Bash

Computational Skills: HPC (Slurm, LSF), protein and interaction language models (LLMs, pLMs/iLMs), interpretable ML, Agentic AI for Biological Applications (Biomni, Co-Scientist), multi-omic and sc-omic network analysis (transcriptomics, epigenomics, proteomics), protein structure modeling and visualization (AlphaFold3, OpenMM, PyMOL, fpocket), processing imaging data (ImageJ/FIJI, fMRI, CNN ResNet finetuning)

Laboratory Skills: Genotyping/PCR, ELISA, flow cytometry, calcium imaging, confocal microscopy, molecular cloning, animal handling (mice, *Drosophila*)

RESEARCH EXPERIENCE

St. Jude Center for Applied Bioinformatics: Summer Research Intern

June 2026-Present

Advised by: Gang Wu and Jeremy Chase Crawford

- Prioritized patient-specific cancer neo-epitopes and variants for a B-ALL cohort and identified patterns associated with relapse and subtypes for future clinical translation.
- Guided agentic AI (Biomni and Co-Scientist) and other LLMs through prompt engineering to explore testable hypothesis and design validation strategies.
- Evaluated AI outputs from Biomni and Co-Scientist for biological accuracy.
- Supervised 2 high school interns in the Research Immersion Program, guiding them through hands-on research.

CMU-Pitt Joint Computational Biology: Graduate Student

Sept 2022-Present

Advised by: Jishnu Das

- Developed SWING, a first-in-class interaction-based vocabulary language model (iLM) to predict peptide:peptide interactions and perturbations collaboratively (Published in Nat. Methods).
- Integrated multi-omic datasets and performed network-based analysis to identify a core T follicular helper (Tfh) cell network that is consistent across contexts (Accepted in Principle at Nat. Commun.).
- Collaborated to identify novel mechanisms of HIV infection in Tfh cells. (In Revision at Nat. Commun.).
- Extended iLMs to capture ternary (pMHC:TCR) and non-peptide partners (small-molecule:protein) interactions, enabling generalizable, patient-specific predictions for clinical translation.

Virginia Tech School of Neuroscience Ni Lab: Laboratory Manager

July 2019-July 2022

Advised by: Lina Ni

- Created an ImageJ plugin to automate data extraction and processing of experiments (Published in JoVE).

- Identified the function, molecular components, and properties in the thermosensory pathway of *Drosophila* larvae (Published in JEB & Front. Mol. Neurosci.).
- Designed automated workflows for fly tracking to ensure reproducible results and speed up data analysis (JEB).
- Generated new fly strains for analysis of gene function through molecular cloning and recombination.

Landos Biopharma: Laboratory Technician

July 2018- July 2019

Advised by: Josep Bassaganya-Riera

- Identified the mechanism of action for drugs through evaluation of differentially expressed gene (DEG) patterns.
- Collected and analyzed data from necropsies, cell cultures, flow cytometry, ELISA, and western blots.
- Handled, monitored, and performed routine medical treatments on mice, rats, and pigs.

NYU Langone Health: Volunteer Researcher

April 2018-June 2018

Advised by: Timothy J. Cardozo

- Designed structure and evaluated stability of antibody eliciting epitopes for staph database and staph vaccines.
- Analyzed alignments to predict protein structures through homology modeling of proteins not in the PDB.

Nathan Kline Institute: Research Assistant

September 2016- May 2018

Advised by: Cameron R. Craddock

- Determined functional similarities of the brain between healthy individuals with comparable personality traits.
- Mapped cluster differences of fMRI data on brain templates to visualize subgroup differences in activation.
- Assessed the quality of 225 skull-stripped brain images and manually edited them.

Child Mind Institute: Volunteer Research Assistant

September 2016-May 2018

Advised by: Steven Giavasis

- Improved the speed, usability, and statistical implementation of Python scripts used by the institute.
- Enhanced a Python script to utilize a Docker container and parallelize the data processing increasing the speed by 30%. The script is included in the current version of C-PAC software.

PUBLICATIONS

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. *bioRxiv*. [Accepted in Principle at Nature Communications]

Del Rio Estrada, P. M.*, Georgakis, S.*, Orfanakis, M., **Omelchenko, A.**, Coirada, F., Dos Santos, J., ... & Ribeiro S.P (2026). IL-10-and TGF- β -Driven BCL6 Expression Suppresses Antiviral Defenses and Renders Lymph Node T Follicular Helper Cells Permissive to HIV Infection. *Preprint*. [In Revision at Nature Communications]

Omelchenko, A. A.*, Siwek, J. C.*, Chhibbar, P.*, Arshad, S., Rosengart, A., Nazarali, I., ... & Das, J. (2025). Sliding Window Interaction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. *Nature Methods*, 1-13.

Omelchenko, A. A., Bai, H., Hussain, S., Tyrrell, J. J., Klein, M., & Ni, L. (2022). TACI: An ImageJ Plugin for 3D Calcium Imaging Analysis. *JoVE (Journal of Visualized Experiments)*, (190), e64953.

Omelchenko, A. A., Bai, H., Spina, E. C., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. (2022). Cool and warm ionotropic receptors control multiple thermotaxes in *Drosophila* larvae. *Frontiers in Molecular Neuroscience*.

Huda, A.*, **Omelchenko, A. A.***, Vaden, T. J.*, Castaneda, A. N., & Ni, L. (2022). Responses of different *Drosophila* species to temperature changes. *Journal of Experimental Biology*, 225(11), jeb243708.

Tyrrell, J. J., Wilbourne, J. T., **Omelchenko, A. A.**, Yoon, J., & Ni, L. (2021). Ionotropic Receptor-dependent cool cells control the transition of temperature preference in *Drosophila* larvae. *PLoS genetics*, 17(4), e1009499. <https://doi.org/10.1371/journal.pgen.1009499>

Wang, X., Li, X. H., Cho, J. W., Russ, B. E., Rajamani, N., **Omelchenko, A.**, Ai, L., Korchmaros, A., Sawiak, S., Benn, A.R., Garcia-Saldivar, P., Wang, Z., Kalin N.H., Schroeder, C.E., Craddock, R.C., Fox, A.S., Evans, A.C., Messinger, A., Milham, M.P., Xu, T. (2021). U-Net Model for Brain Extraction: Trained on Humans for Transfer to Non-human Primates. *NeuroImage*, 118001. <https://doi.org/10.1016/j.neuroimage.2021.118001>

*These authors contributed equally

CONFERENCES and AWARDS

Presentations:

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: AAI, April 2026.

Computational approaches to understand immune system modulation through protein interactions. Sip of Science, December 2025.

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: Center for Systems Immunology Retreat, November 2025

A unified network systems approach uncovers a core novel program underlying Tfh differentiation. Oral Presentation at: ISMB/ECCB, July 2025.

Sliding Window Interaction Grammar (SWING): a generalized interaction language model for peptide and protein interactions, Oral Presentation at: Graduate Research Symposium, November 2024

Sliding Window Interaction Grammar (SWING): a generalized interaction language model for peptide and protein interactions. Oral Presentation at: ISCB GLBio, May 2024

SWING- A generalizable language model for protein and peptide interactions, Oral Presentation at: Center for Systems Immunology Retreat, September 2023

Ionotropic Receptor-dependent warm cells in *Drosophila* larvae, Oral Presentation at: VT School of Neuroscience: Summer Research Retreat, August 2020

Posters:

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 25th Annual Immunology Retreat, October 2025.

Omelchenko, A. A., Rahman, S. A., Viswanadham, V. V., Yuen, G. J., Estrada, P. M. D. R., D'Onofrio, V., ... & Das, J. (2025). A unified network systems approach uncovers a core novel program underlying T follicular helper cell differentiation. Poster presented at: 23rd Annual Immunology Retreat, October 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: Graduate Research Symposium, November 2023.

Omelchenko A. A., Siwek, J., Chhibbar, P., Rosengart A., Koes D., Joglekar A., Das, J., A generalized language model for predicting perturbations of protein-protein and MHC:Peptide interactions. Poster presented at: 21st Annual Immunology Retreat, October 2023.

Omelchenko, A. A., Tyrrell, J. J., Wilbourne, J. T., & Ni, L. Ionotropic Receptor-dependent dorsal organ warm cells mediate warm sensing in *Drosophila* larvae. Poster presented at: SfN, November 2021; Virtual

Tyrrell, J. J., Wilbourne, J. T., Omelchenko, A. A., Yoon, J., & Ni, L. Ionotropic Receptor-dependent cool cells control the transition of temperature preference in flies. Poster presented at: VT Molecular and Cellular Biology Summer Event, August 2021; Blacksburg, Virginia.

Honors and Awards:

Travel Awards:

St. Jude Travel Award for BioHackathon participation
BGSA Travel Award for Abstract at ISMB 2025 Conference and Talk
CRA-WP Grad Cohort For Women Workshop Travel Award 2023

Presentation Awards:

AAI Trainee Abstract Award- AAI 2026
Most Technically Impressive Project – St. Jude Hackathon 2025
Best Abstract – Center for Systems Immunology 2025
Best Abstract – Graduate Research Symposium 2024
Best Talk – CPCB Retreat 2024
Best Abstract – Center for Systems Immunology 2023

Academic Awards and Scholarships:

Dean's List- Awarded for academic achievement 3 years in a row 2015-2018
Trio Scholar—2013-2018
STEM Women in Engineering Scholarship—2013-2018

LEADERSHIP AND SERVICE

Pennsylvania Woman Work Mentor – Mentoring individuals who are ready for their next career step and need guidance entering the workforce. July 2025-Present

St. Jude HSRI Mentor – Mentored 2 high school interns in the Research Immersion Program. June 2026-Present

Allegheny County Science Policy Member- Engage in science policy, advocacy, and communication at a local, state, and national level. May 2026-Present

BGSA CPCB Program Representative – Represent the CPCB graduate program in the School of Medicine student government. September 2023-Present

CPCB Steering Committee- Graduate representative from the CPCB program chosen to meet with the directors across CMU and PITT to steer the progress of the program. September 2024-September 2025

CPCB GSA Liaison – Connected the CPCB graduate program to the larger network of the School of Medicine student government. September 2023-July 2025

TechBIO REU mentor – Advised undergraduate students in research ethics as part of the TechBIO REU program. June 2023-August 2023, June 2024-August 2024, June 2025-July 2025

Book Club Mentor - Guided first-year students and facilitated their transition into the graduate program through discussions of assigned readings. September 2023-May 2025

First-Year Ambassador Mentor - Provided assistance and support to incoming students to assist in their acclimation to the graduate school. September 2023-2024

TRAINING

Teaching:

Teaching Assistant. Computational Structural Biology, University of Pittsburgh,

Sept 2023-Dec 2023

- Developed curriculum, lesson plans, and course materials and led weekly recitations.
- Graded assignments and final projects and held weekly office hours.

Guest Lectures and Invited Talks:

"A Brief Overview of Computational Structural Biology," guest lecture, Data to Knowledge course, University of Pittsburgh, June 2023.

"How to Network at GLBio," invited talk, ISCB GLBio Workshop for Early Career Researchers, May 2024.

Relevant Graduate Coursework:

- Machine Learning for Biomedical Applications (PITT)
- 10-701 Introduction to Machine Learning (CMU)
- Genomics (CMU)
- Cellular and Systems Modeling (PITT)
- Computational Structural Biology (PITT)
- Essential Mathematics and Statistics for Scientists (CMU)

- Computation for Data Science I (Virginia Tech)
- Problem Solving in Genetics, Bioinformatics, and Computational Biology (Virginia Tech)
- Computer aided Drug design (NYU)
- Protein Engineering (NYU)
- Biocatalysts (NYU)
- Biosensors and Biochips (NYU)
- Immunology (NYU)